REQWEST: A BARANGAY DOCUMENT REQUEST SYSTEM FOR THE RESIDENTS OF WEST KAMIAS, QUEZON CITY

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

Barangays are the Philippines' smallest governmental units, and they contribute to ensure the safety of its citizens. Barangays are also in charge of providing and disseminating vital official papers, such as clearances and certificates, as requested by citizens. Taruc et al. (2023) found that barangays that rely on manual document request processing face several issues, including difficulties in managing files and records, ensuring file security due to the absence of a secure repository, and generating barangay records in real-time.

In many cases, barangay residents have a hard time processing and requesting their documents in their barangay. Some are having trouble queueing a long line and residents being confused on the requirements needed. Common issues include delayed responses from barangay officials, difficulties in managing and retrieving files, and a lack of organized record-keeping.

According to researchers from Olongapo City University (2021) Residents in various barangays have reported challenges in obtaining necessary documents due to inefficiencies in manual processing systems.

Having a Web-based Document request system can help the barangay administrators to track and compile the requirements of the residents and help the residents to have their request submitted online and waiting for them to pick it up.

REQWEST are designed to alleviate the difficulties residents face in obtaining necessary documents, promoting a more efficient and responsive barangay administration.

Theoretical Framework

The Barangay Document Request System aims to enhance the overall accessibility, efficiency, convenience and transparency of public services that are offered by barangays. According to Maribao, Delito, Cos, and Monticalvo (2024), the integration of technology into public service systems can solve issues like long processing times, document request errors, and challenges in required information.

This study investigates factors such as the type of document requested (i.e., Barangay Clearance, Certificate of Indigency, Business permits, Certificate of Residency), the effectiveness of the system in streamlining the document request process, and the effect of digitalization on waiting time, rejection rates, and the overall satisfaction of both barangay staff and constituents. The Technology Acceptance Model (Davis, 1989) has explained the relationship between these variables, with the focus on ease of use and functionality of the technology in the adoption process. In the context of this study, the Barangay Document Request System facilitates efficient document requests and the tracking of its status, providing proper communication between the barangay and its residents.

The key factors that influence technology adoption include system accessibility, ease of navigation, absence of long waiting times, and user satisfaction. It is acknowledged that the preparedness and willingness to embrace new technologies may be different among both the barangay staff and the residents. The successful adoption of the system depends on effective communication, training and emphasizing the advantages of converting from a physical to digital system for document requests. The study will assess the system's usability and its effects on service delivery, with the purpose of contributing to a more convenient and user-centered approach to barangay administration.

Conceptual Framework

INPUT

- User Information (name, address, contract details, priority group)
- Appointment Schedule (date & time)
- Type of Document (barangay clearance, indigency, business permit, renewal of business permit)

Hardware Requirements

- Laptop/PC
- Mobile Device

Software Requirements

- PHP
- HTML, CSS, JavaScript

PROCESS

- Barangay Document Request for the citizen of Barangay West Kamias

System Modules

- Login Module
- Request or Appointment Module
- Chatbot Module
- Profile Module

OUTPUT

- Claiming Schedule
- Application Status
- Notification and Feedback
- Frequently Asked Questions (FAQs)

Admin Side

- Application Information
- Administration Dashboard
- List of Applications

Statement of the Problem

This study aims to address the challenges faced by the barangay community regarding the requesting of necessary documents, such as barangay clearances and certificates of residency. Moreover, this sought to answer the following questions:

1. How can a digital document request system improve the efficiency of barangay services?
2. What security measures should be implemented to protect residents' data and privacy?

3. How can the system improve transparency in the request and approval process?
4. How can the system ensure accessibility for residents with limited technical knowledge?
5. How effective is the REQWEST: Barangay Document System for Residents of West Kamias, Quezon City in addressing the following aspects:

    5.1 Request processing time

    5.2 Accuracy of document issuance

    5.3 User-friendliness of the interface

    5.4 Accessibility for all residents

    5.5 Data security and privacy; and

    5.6 Transparency in the request process

Scope and Limitation of the Study

The researchers are working on a centralized web-based system that will facilitate document requests such as Barangay clearances and certificates of residency. The system will allow residents to submit the requests online, track the status of their applications, and directly receive notifications on updates or completions.

However, The system will be limited to Barangay Kamias, Quezon City, and will not support document requests from other areas. Furthermore, similar to payment processing or integration with other local government systems beyond the barangay's database. Manual processes such as document approvals may still require barangay personnel to process tasks that cannot be automated via the system.

The REQWEST operates through three (3) authorized system users: barangay chairman, secretary, and local residents. The system only includes the following:

1. A system-based document authentication cannot process any requests for documents or complaints from unregistered residents.
2. Individuals who are not capable of using the internet will not access barangay online functionality, so individuals must submit their requests by visiting the barangay office physically.
3. Only users that are authorized by the system administrator can access the system.
4. Barangay Certificates and Certificates of Indigency are the only documents that can be requested and only authenticated documents can be processed once it is checked from the system's database.
5. When an issue is reported, only the photos given by the residents are acceptable as supporting evidence.
6. Since the system is still in the developing stage, it would not include highly technical statistics such as online payment transactions, which could be put forward in the recommendation for future studies.

Significance of the Study

This study would be beneficial to the following:

1. Barangay Officials – Officials of Barangay West Kamias will benefit from having better access to and management of resident information and document requests. The system will streamline administrative processes, improving efficiency in handling service requests.

2. Citizen (Brgy. West Kamais) – Citizens will have easier access to request documents such as birth certificates (PSA), barangay clearances, voter's IDs, barangay IDs, indigency certificates, and business permits. The system also prioritizes certain groups, including:

   2.1 Senior Citizen and Person with Disability (PWD) - These individuals will receive priority service and free service as stated by the barangay secretary.

   2.2 Student – Students also receive free services.

3. Future Researchers – Future researchers will benefit from this study as it provides a foundation for further development. They can improve and refine the existing system to address emerging needs and enhance its functionality.

Definition of Terms

Document Management System. This term refers to a system that tracks and stores documents digitally to enable easier access and retrieval.

Efficiency. The ability of the system to achieve maximum productivity with minimal effort.

Functionality. This term refers to the features of a system, describing what it can do and how it performs various tasks.

Usability. This term refers to the ease with how users can interact with the system to achieve their goals.

Reliability. The consistency of a system, ensuring it performs its intended functions without failure.

Security. This term refers to the measures and protocols put in place to protect data and systems from unauthorized access.

System Implementation. The process of installing and integrating a system into an operational environment, such as local barangays.

Public Service. Services provided by the government to meet the needs of the public.

Community Engagement. The process of involving community members in planning, and execution of projects or services.

Data Management. The processes used to collect, store, and organize data for accessibility.

Web-based System. A system or application that runs on a web browser, accessible via the internet.

Priority Groups. Groups that are given priority in receiving services or benefits due to specific needs, such as senior citizens or persons with disabilities (PWDs).

User Interface. A digital surface in which a user interacts with a computer or software, including the layout, buttons, and controls on a screen.

Transparency. The openness and clarity of actions, processes, or decisions within a system.

Record-Keeping. The practice of systematically maintaining and storing records for reference, legal, or administrative purposes.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

To validate the use of a web-based portal as a potential document request system, the following literatures have been provided to further expand the knowledge and explain the purpose as to why the study will be conducted. Only those that were relevant were studied.

Barangay Information Management System

As the smallest political unit in the Philippines' governmental structure, the barangay is responsible for the fundamental platform for local administration. Most barangay operations today depend on manual techniques for record management beside information processing. Taruc et al. (2023) found that barangays that rely on manual document request processing face several issues, including difficulties in managing files and records, ensuring file security due to the absence of a secure repository, and generating barangay records in real-time. Clearances and certifications are prepared with less to no base-line data to assure reliability. This opens the risk for nonresidents to take due advantage of this process flaw (Ong, 2020). As stated by Arcega et. al (2023), Barangay administration has document and record management as an essential duty since these records support both legal documents and certification needs as well as residency verification. The essential management of documents and records are available to facilitate various purposes such as barangay clearance and certificate generation alongside ID production and proof of residency and other barangay documents.

Barangay Management Information Systems now dominate as people depend more intensely on computer systems across all aspects of life, especially in business operations. By guiding business day-to-day operations, the productivity of workers increases because they spend less time on data collection and more time addressing essential work functions. (Maribao et. al, 2024)

Accordingly, residents formerly needed to visit the barangay hall, complete paperwork, and then have to wait extended processing time to obtain these documents from the local office. The development of technology together with growing public service effectiveness demands has led numerous barangays to implement the Barangay Document Request System.

In a study conducted by Santiago et al. (2023), With technological advancements and the growing demand for more efficient public services, many barangays have started to implement digital solutions to simplify administrative processes. One such innovation is the Barangay Document Request System, which allows residents to request and receive barangay documents online. This system greatly reduces processing time, enhances accessibility, and improves the overall efficiency of barangay operations, effectively addressing the challenges posed by traditional methods. Castagna and Bigelow (2021) define information technology as the development, processing, storage, protection, and sharing of electronic data through computers, storage, networking, and other physical devices. This move towards digitalization highlights the increasing significance of using technology to deliver timely and effective public services, in line with the broader objective of enhancing governance and community engagement at the local level.

Web-based Appointment System

According to Maribao et al. (2024), the Barangay Management System (BMS), was established to invigorate governance and provide timely services to barangays, in their view, is the most basic unit of governance in the Philippines. This system comprises various features, including those pertaining to a resident database, record-keeping, financial control, service requests, and event organization. Such features are designed to enhance workflow within administration, reduce errors, and provide efficiency in the operation of the barangay in a couple of concerns.

It explains that there are many advantages to the automated implementation of barangay processes: increased transparency and accountability, and data-backed decision-making. Thus, the system allows barangay officials to make informed decisions since data is very accurate and up-to-date and supports an evidence-based municipal governance model. There is increased interaction with the community through event management and back-and-forth engagement with citizens.

This research establishes that the integration of technology into barangay operations needs to be established not only to address inefficiencies but to further widen the scope for improving service delivery and community engagement.

Nagrama et al. (2024) stated that residents had to visit the barangay hall in person to request these documents. This process required them to fill out forms manually and wait for their requests to be processed, often leading to delays and inconvenience. However, with technological advancements and the growing demand for more efficient public services, many barangays have started to implement digital solutions to simplify administrative processes. One such innovation is the Barangay

Document Request System, which allows residents to request and receive barangay documents online. This system greatly reduces processing time, enhances accessibility, and improves the overall efficiency of barangay operations, effectively addressing the challenges posed by traditional methods.

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According to Ballaran, Donaire, Singzon, Naz, Tanael, and Centeno (2023), the Development and Evaluation of a Web-Based Barangay Profiling and Issuance System aimed to address inefficiencies in barangay governance by leveraging a centralized, web-based platform. The system aimed to make the gathering, storage, and issuance of barangay certificates and permits simpler. Regression analysis illustrates the way technology exponentiates service delivery and sharpening of decision-making processes.

Among the major cases were the lack of centralized databases in barangays, wrong data presented by the residents, and lack of proper allocation of resources. This system was developed to eradicate these problems by allowing secure data management and document processing with proper resource allocation. Evaluations apparently showed a great improvement in functional suitability, performance efficiency, and reliability of the system, which makes the platform a powerful tool in improving transparency and accountability in barangay governance. This study brings into focus the imperative need for integrating the adoption of digital solutions toward modernizing the operations of local governments and enhancing the quality of services to their constituents.

AI Chatbot

According to Pranav Nataraj Devaraj et. Al (2024), managing legal documents can be difficult due to their complexity and size. They developed an AI chatbot to make the process easier. The chatbot uses Langchain to break down large texts and find relevant information, a Flask backend to handle queries, and an Android app for user interaction. This system helps users, even those without legal expertise, access and understand legal documents more easily. The authors aim to improve it further by adding features like better design, support for more documents, and higher query limits. By responding to inquiries, providing clarifications, and supplying further resources, AI chatbots can offer prompt assistance. Additionally, chatbots can support teachers in a variety of ways by serving as virtual teaching assistants. This research aims to comprehend the whole advantages of AI chatbots in education, as well as their potential drawbacks, chances, difficulties, and future prospects (Labadze, Grigolia and Machaidze, 2024).

According to Xueliang Zhao et. Al (2019), selecting the right response in retrieval-based chatbots is challenging, especially when conversations rely on external knowledge like documents. To address this, they developed a Document-Grounded Matching Network (DGMN) that matches conversation contexts and candidate responses while grounding them in relevant documents. The model uses layers for encoding, fusing, and matching information to dynamically identify important details from both the context and the document. This approach significantly improves chatbot performance on datasets like PERSONA-CHAT and CMUDoG, outperforming previous methods and demonstrating how integrating external knowledge enhances conversational AI.

Synthesis of the Reviewed Literature and Studies

The studies highlight the growing importance of technology in local barangays. Traditional processes in barangays, such as document management and requests, are filled with efficiencies, including delays and mismanagement. These issues hinder the accessibility of essential services that the barangay should be providing.

Digital innovations such as web-based portal document request systems can address these challenges by improving efficiency and transparency. The system uses automated processing and secured data management to reduce errors and enhance its current service.

AI-powered tools like chatbots can enhance the service by integrating the bots to the current process. This can enable efficient access to information and improve user interaction. Chatbot paired with the document management system should align with the goal of modernizing the local government to meet increasing demands for effective public service delivery.

The synthesis emphasizes that integrating technologies into local barangays not only addresses efficiency but also improves the traditional system to deliver timely, accurate and citizen-focused services.

Chapter 3

METHODOLOGY

Research Design

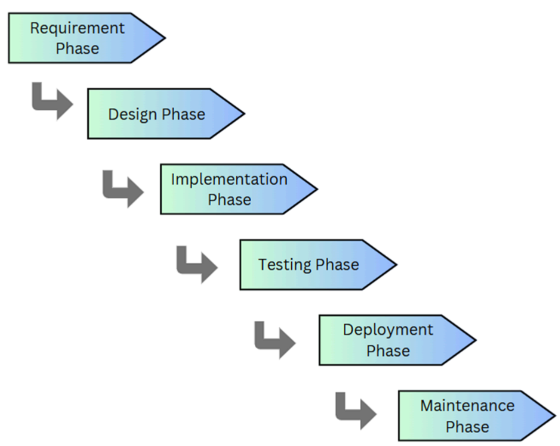
A thorough plan for the barangay web-based document request system was created during the system design phase. It is developed using the specifications acquired in the previous stage. This scheme contains a number of crucial features, including a program design, logical flow diagram, design, as well as the actual design and specs. According to the data flow diagram, the flow of data in the system, whereas the software design describes the general framework of the program. Furthermore, the logical design distinguishes the various parts of the system, including the database and user interface, and how they work together. Conversely, the hardware and software are defined by the physical design and specifications. prerequisites for system implementation. When combined, these elements offer a thorough plan for creating the barangay barangay web-based document request system, making sure it complies with the requirements of end users while abiding by best practices and industry norms.

Software Development Methodology

During the implementation phase, developers will produce a system aligned with the specifications and requirements defined in the design phase. Each system component will be developed consecutively, ensuring adherence to the initial architecture and functionality. Following this, the system will move to the testing phase, where rigorous evaluations and testing will be done. These tests will ensure that the system delivers the desired performance and service as mentioned in the requirements phase.

Once testing is complete, the system will enter the deployment phase, where it will be implemented in the barangay for end-users to make use of. Feedback will be collected to assess its initial performance.

Figure 2. Waterfall Methodology



The maintenance phase will then begin, during which developers will continue to monitor the system's performance, identify and resolve technical issues, and use the feedback gained from end-users. Improvements will be applied based on the feedback to remain an effective and user-friendly system until the system fully satisfies the need of the users.

Data Case Analysis

Our research adopts a descriptive mixed-method approach to develop the Barangay Document Request System. We gathered data from two sources to evaluate the effectiveness and efficiency of the proposed system.

The first source is primary data, which was collected through surveys distributed to barangay residents and officials. These surveys identified key challenges in the current document request process, such as delays, inefficiencies, and pain points during documentation. This primary data serves as a valuable resource in designing and building the system.

The second source is secondary data, which includes historical records, studies, and literature about barangay document requests and chatbots. This data provides a benchmark for comparing the performance of the current system with the expected improvements after implementing the new digital system.

By integrating both primary and secondary data, this study offers a thorough analysis of the existing problems and the potential benefits of a streamlined, digital solution. This approach ensures that our research is both comprehensive and reliable, combining real-world data with insights from previous studies to enhance the system's design and implementation.

Sources of Data

The researchers gathered information by interviewing barangay officials in West Kamias. The researchers ask in-depth questions about barangay forms, documents, and proper services. Furthermore, the researchers supplied a consent letter to the participants to make sure they were aware of the 10-item questionnaire that talks about the difficulties in getting documents in their local barangay.

Research Instrument

The researchers will conduct an interview supported by a guided questionnaire, which will be used to gather essential information to aid in the development of the proposed system. The questionnaire consists of 10 questions related to barangay documents, forms, and other common services needed for daily office operations. Participants will be asked to provide factual and accurate responses, ensuring that the collected data is reliable and useful for the system's design.

Data Generation Procedure

The purpose of this study is to design a simpler and more efficient barangay document request procedure in Barangay West Kamias, Quezon City. For this, the researchers designed the REQWEST system, a system to modernize and simplify this process, by making it simpler, more accurate, and easier for all. The structured approach the researchers took to collect data was used to gather the information needed to build this system.

Surveys were conducted for residents and interviews with the barangay officials in West Kamias, the researchers combined the two types of inputs to come up with insights. The surveys were strategically designed to seek to uncover the particular needs, difficulties, and patriarchy of residents in asking for documents. Interviews with barangay officials, however, the struggles faced, and what they require to make the system more effective.

The data collection focused on two main groups, the residents of Barangay West Kamias and the barangay officials. The survey included 100 residents, enough for a representation of a diverse population. Barangay staff on the other hand, namely, the barangay secretary, and chairperson were interviewed to know how things are currently being done.

The survey was distributed both online and offline to make sure anyone could participate. To have more meaningful conversations, the researchers spoke with barangay officials personally for the interviews. All the responses were noted carefully and analyzed and common issues and areas where improvements can be made are identified. That's the foundation for designing the REQWEST system that will help both the residents and the barangay staff make life easier.

Ethical Considerations

In this research, the data collected by face-to-face interview and online survey was conducted to the citizens of said barangay. These methods are mainly focused on how many average daily numbers of citizens are requesting documents, and a survey that shares the highest daily request in the barangay.

This research is centralized to the citizens of Barangay West Kamias District 3, Quezon City. Having nearly 4,000 residents, the sample size will be around 100 with a margin of error of 10%. Initiating this research to the barangay, researchers sought for authorization to the barangay captain. The researchers have adhered to the ethical principles of informed consent and respect for participants' privacy. Formal intent letter was made by the researchers that is passed and shown to the participants, outlining the study's purpose, methodology, and scope. Additionally, researchers have provided a valid I.D document to verify their identities and credentials, assuring the uniqueness and legality of the research. Setting the compliance with ethical and professional standards is crucial for ensuring the validity, reliability, and generalizability of the research findings. We informed the interviewee that the purpose of our study is to provide e-services to the barangay residents, focusing on addressing their document-related concerns. The study aims to give immense convenience to the residents, especially with the growing population, by reducing the need for them to wait in long queues daily.